

A Mini-Project Report
On
**"PORTABLE VENTILATOR USING ARDUINO
PROGRAMMING"**

SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF DEGREE OF
BACHELOR OF ENGINEERING
IN
ELECTRONICS & COMMUNICATION ENGINEERING

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Certificate

This is to certify that the Mini-project report entitled “**PORTABLE VENTILATOR USING ARDUINO PROGRAMMING**” is being submitted by **Mr. Dr. I SHARATH CHANDRA, Ms. KHATIJAMAHVEEN(1608-20-735-084), Mr. DEVARAKONDA SUDHA SAI KRISHNA (1608-20-735-070), Mr. ETHAMUKKLA SUPRATEEK(1608-20-735-083)** in partial fulfillment for the award of the Degree of Bachelor of Engineering in Electronics and Communication Engineering of the Osmania University, Hyderabad, during 2022-23, is a record of bonafide work carried out under our guidance and supervision.

The results presented in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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DECLARATION

This is to declare that the work submitted in the present Mini- project work report titled **“PORTABLE VENTILATOR USING A”** is a record of bonafide work done by me in the Department of Electronics & Communication Engineering, Matrusri Engineering College, Saidabad, Hyderabad.

No part of the report is copied from books, journals, internet and wherever the subject content is taken; the same has been duly referred in the text. The report generated is based on the project work carried out entirely by us and not copied from any other source.

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We would like to take this opportunity to place it on the record, that this Mini-project would never have taken shape but for the cooperation extended to us by certain individuals. Though this is not possible to name all of them, it would be a pardonable on us part if we don't mention some of the very important persons Sincerely we acknowledge our deep sense of gratitude to the project guide, **Dr. I SHARATH CHANDRA**, Assistant Professor for his constant encouragement, help and valuable suggestions. we wish to thank him for his constant motivation and help throughout the Mini-project.

We would like to express our deep gratitude to the Head of Department **Dr. N SRINIVASA RAO** and Project Coordinators **DR. PALLAVI KHARE, ALEDDULA ABHISHEK REDDY** for their timely cooperation while carrying the Mini-project. It is their friendliness that made me feel free and learn more from them.

Last but not the least we would thank to all those people associated directly or indirectly with the Mini-project.

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ABSTRACT

Mechanical ventilation is a life-saver in the development of modern ICU'S. Thus our ventilator is inspired by the beauty of simplicity, this way we have made the technology much cheaper and less expensive to make and maintain. This kind of ventilators can be used in low-income countries as respiratory failure complicates most critically ill patients.

This is proposed to have better working performance than already available in market at a very low cost, it's also rapidly deployable ventilator. The benefit here is its standard components are accessible to public, open source compatibility to allow anyone to recreate the system, ability to ventilate in acute problems, provide adequate oxygenation and ventilation, simple design appears to be safe and allow rapid mass production if ventilator surge demand exceeds the supply.

This is completely off label use, as it serves as a bridge device and helps with its availability. This may allow less severe patients to be cared, while resources are focused on those most in need. In conclusion, I would like to say this can be built quickly and affordably with its ability to respond to pressure changes in patients and control how fast oxygen is to be delivered.

Keywords: Mechanical ventilator; low-cost ventilator; COVID-19, artificial ventilation; health monitoring; Bag Valve Mask (BVM), Portable and Automatic.

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Chapter 1

1.1 Introduction

A ventilator is a machine that provides mechanical ventilation by moving breathable air into and out of the lungs, to deliver breaths to a patient who is physically unable to breathe or breathing insufficiently. The patient is connected to the ventilator with a hollow tube (artificial airway) that goes in their mouth and down into their main airway or trachea. They remain on the ventilator until they improve enough to breathe on their own. Almost every bed in a hospital has a manual resuscitator nearby, available in the event of a rapid response or code where healthcare workers maintain oxygenation by squeezing the bag. Automating this appears to be the simplest strategy that satisfies the need for low-cost mechanical ventilation, with the ability to be rapidly manufactured in large quantities. However, doing this safely is not

trivial. For this reason, we designed Smart portable bio- mechanical patient monitoring ventilator it is a concept to realize the advancement in health monitoring system of human beings due to the COVID-19 pandemic, the medical facilities have been scares and are required to a lot of people.

□ Respiratory diseases and it's failure constitute a major public health problem in both developed and less developed countries. Asthma, chronic obstructive pulmonary diseases are widespread.

□ These conditions are instilled by air pollution, smoking, etc all of which are on the rise in developing countries. Patients with underlying lung disease may develop respiratory failure and hence they can be supported by mechanical ventilation .

□ Providing ventilator support to an ICU patient as it serves as a bridge device and help with its availability. This may allow less severe patients to be cared ,while resources are focused on those who are most in need .

As global demand for Ventilators is increasing and there is shortage of Ventilators in our country as well, also managing patients during pandemic time is big task, battery operated portable Ventilator using AMBU (Artificial Manual Breathing Unit) bag is developed which supply oxygen to the patients and measures oxygen saturation (SPO2) of

patients and visible on LCD Display.

1.2 Problem specification

- Severely affected patients from respiratory problems need Ventilators, we chart the challenges faced by the health care systems in procuring these machines and the role of ambu bag in building ventilators.
- The seriously ill patients need respiratory support, as their lungs get damaged by the coronavirus leading to breathing difficulties. Ventilators are needed in such cases for supplying adequate oxygen (O₂) into their lungs and remove carbon dioxide as life saviour.

1.3 Objectives

- To design a portable mechanical ventilator by automating the operation of ‘ Ambu Bag‘ with volume control ventilation which can be used in respiration.
- The Ventilator delivers breaths by comprising a conventional Bag-Valve Mask (BVM) using Servomotors and Microcontroller with coding
- An easy to use life-sustainer that requires no specially train staff. So, our main objective of making this project is to overcome most and all the problems that occurs in all critical condition for saving lives.

1.4 Methodology

The figure represented the proposed block diagram of the project. It consists of microcontroller where we are using Arduino UNO which acts as master and slave we dumped the code in both the controller in case of one gets failure other gets executed and using AMBU BAG the basic mechanism of working of ventilator start supplying the oxygen to the patient with the control of pressure, volume intake and Inhale-Exhale ratio. UV sterilization takes places during the oxygen supply.

High Torque Servo motor is of 270 degrees rotation mechanical motor to press the BVM for producing the oxygen. The Main aim of the controller is to manage the torque, pressure and speed of the motor to provide a proper - parameters for pressing the BVM. The Patient Monitoring System includes Heart beat monitoring, SpO2 ,microcontroller connected with the sensors to provide accurate readings to take immediate action in case of any emergency and it can be constantly monitored by doctors or family members . . In LCD display volume, pressure, BVM can be seen. A backup power supply is made to operate the mechanism in case of any power surge to avoid the failure of the mechanism. We also added for external use Built-in Nebulizer to provide a proper nebulizer for the patients using piezoelectric material, in the container added liquid medicine that gets vaporized and it connects to inlet of BVM

The Fabrication of our project is designed using Solid edge V19 software we use 6mm acrylic board as it is available at low cost and with better strength. electric material, in the container added liquid medicine that gets vaporized and it connects to inlet of BVM .

Additionally a crisis ringer alert is fitted inside the framework to sound a ready when any abnormality is identified. The whole framework is driven by Arduino regulator to appreciate wanted results and to help patients in COVID pandemic and other crisis circumstances.

1.5 Layout of the thesis

The layout of the project explains the ““PORTABLE VENTILATOR USING ARDUINO””. The organization of the project is as follows.

Chapter-1: Introduction

This chapter contains introduction, objectives, problem specification, methodology, contribution of the project.

Chapter-2: Literature Survey

This chapter examines several research studies that employ various machine learning and deep learning methodologies.

Chapter-3: Software / Hardware Implementation

The chapter contains the hardware and software implementation which is used in this project.

Chapter-4 Results

The chapter consists of results of the project and the performance of the automatic plant watering system using Arduino is discussed.

Chapter-5: Conclusion

The chapter contains conclusion and scope for future.

Chapter 2

LITERATURE SURVEY

This text shows the improvement of low-cost, open-source automatic ventilators. This text also shows the numerical method for monitoring patients' pulmonary conditions. With the assistance of a pressure sensor, we will classify whether the patients are healthy or unhealthy lungs. An Arduino board collects the data from the pressure sensor and sends them to a raspberry pi. The raspberry pi commands the acuter and breathing bag compress accordingly. in line with the manufacturer, the pressure sensor can measure differential pressure of up to 70 cm H₂O.

the gear was attached to the servo meter rod. The rod was made of a Plexi glass bar. The radius of this gear is 2.5cm. Aliaksei Petsiuk, Nagendra G. Tanikella, Samantha Dertinger,Adam Pringle, Shane Oberloier, Joshua M. Pearce, (Partially RepRapable automated open-source bag valve mask-based ventilator)[1]

This text shows the event of a straightforward and easy-to-build portable automated mask value bag. This handle with Arduino controller with the real-time package installed on largely rewrap

3d printable parameter component-based structure. For Arduino extensively grows the conceivable outcomes of the controller. A real-time software system gives fundamental capacities to software tasks, like planning, dispatching, inter-task communication, and synchronization Couchman, B. A. et al. (Nurses role in prevention and management of mechanical ventilation related complications) (2006)[2]

In their article titled, medical care of the mechanically ventilated patient: What does the evidence say? Summarized as medical aid and management of mechanically ventilated patients are challenging and requires nursing expertise for knowing the technological issues undying the patient-centered approach. Mechanical ventilation precipitates several actual and potential complications for critically ill clients. The employment of ventilator care is effective in mechanically ventilated patients producing positive outcomes which carries with it four interventions, those are; elevation of the top of the bed, sedation vacation, peptic ulceration prophylaxis, and deep vein thrombosis prophylaxis. The medical aid practice lacks supportive significant evidence for proving one care approach is healthier than the opposite. within the care of mechanically ventilated patients, the simplest medical aid practice is that the use of the evidence-based practice in conjunction with comprehensive and systematic patients. [3]

This paper describes the look and prototyping of a low-cost portable mechanical ventilator to be used in mass casualty cases and resource-poor environments. The ventilator delivers breaths by compressing a traditional Ambu bag with a pivoting cam arm, eliminating the necessity for an individual's operator for the Ambu bag. An initial prototype is driven by an electrical motor powered by a 12 VDC battery and features an adjustable tidal volume up to a maximum of 750 ml. Tidal volume and number of breaths per minute is set as per the default conditions. Future iterations of the device will include a controllable inspiration to expiration time ratio, a pressure escape cock, an LCD screen, and an alarm to point over-pressurization of the system. Through this prototype, the strategy of automated Ambu bag compression is proven to be a viable choice to achieve low-cost, low-power portable. [4]

Chapter 3

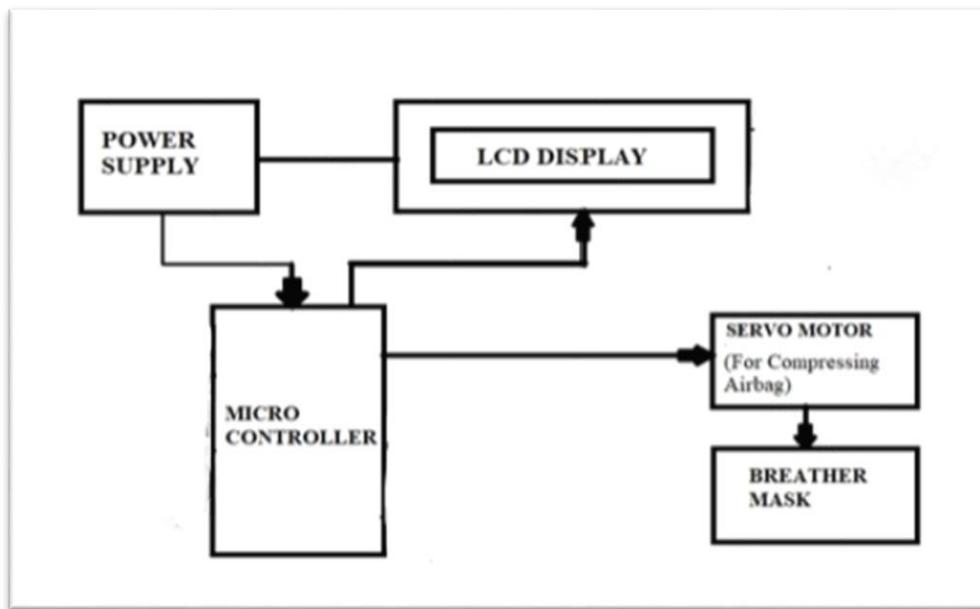
HARD WARE/ SOFTWARE IMPLEMENTAION

3.1 Introduction

Human lungs utilize the opposite pressure produced by the compression movement of the stomach to suck in air for relaxing. An incongruous movement is utilized by a

ventilator to swell the lungs by siphoning type movement. A ventilator component should have the option to convey inside the scope of 10 30 breaths each moment, with the adaptability to manage rising augmentations in sets of two. along with this, the ventilator should have the ability to manage the air volume drove into the lungs with every breath. Last however presently least is that the setting to control the time length for inward breath to exhalation proportion. Aside from this the ventilator should have the option to screen the patient's blood oxygen level and breathed out lung strain to keep away from over/under gas tension at the same time. The ventilator we here plan and foster utilizing Arduino envelops of these prerequisites to create a solid yet reasonable DIY ventilator to aid seasons of pandemic. The whole framework is driven by an Arduino regulator to acknowledge wanted results and to help patients inside the COVID pandemic and other crisis circumstances.. As creators, it's our obligation to battle the deficiency by building improvised open-source substitute gadgets. One significant gadget that request has inclined up is ventilators for patients who need help with their breathing thanks to the respiratory impacts of COVID-19. Fundamentally, a ventilator could be a machine that gives breathable air into and out of the lungs, to convey breaths to a truly incapable patient to inhale, or breathing deficiently. A DIY ventilator probably won't be proficient as that of a clinical grade ventilator yet it can go about as a fair substitute on the off chance that it's command over the ensuing key boundaries

Block diagram and individual explanation of blocks

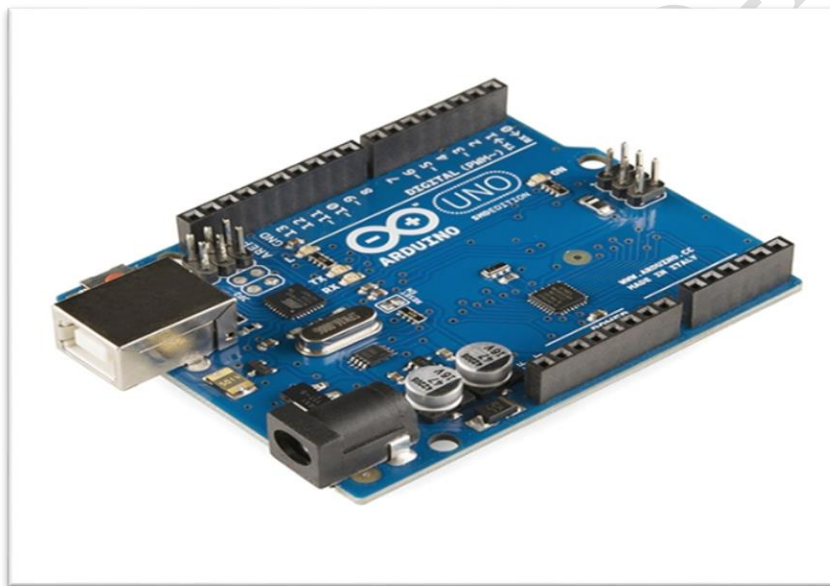


Block diagram of the ventilator

The diagram of a ventilator using Arduino with is shown within the above Fig. Rapid prototyping technologies were accustomed create a medical ventilator. The unnatural physical respire element is joined to the wall oxygen source employing a flow meter as an air reservoir.

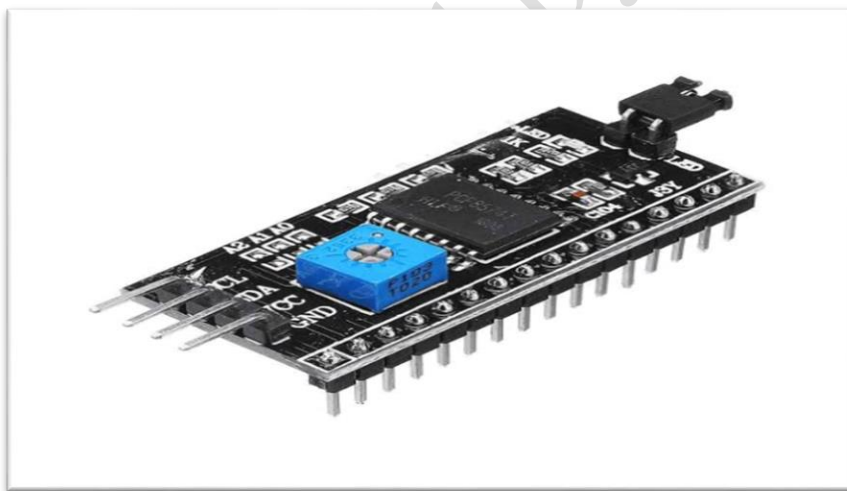
Individual blocks explanation:

Arduino UNO



The Arduino Uno is an ATmega328 grounded microcontroller. It features 14 digital I/O legs, among which 6 can be used as PWM labors, the rest of the legs include 6 analog inputs, a 16MHZ demitasse oscillator leg, power jack point, USB connection harborage, an ISCP title leg, and a reset button. It can be powered either by with a USB string or with an AC- to- DC attachment or a battery. Though this board can accept voltages between 7 to 20 V, its operating voltage is 5V. This board can be programmed using an open-source software tool Arduino IDE.

I2C module:



This is a RoHS compliant I2C Serial LCD Daughter board that can be connected to a standard 16×2 or 20×4 Character Display Module that supports 4-bit mode. All Character Modules sold on our site support 4-bit mode, and nearly all commercially available 16×2 and 20×4 line character modules support it too. This board has a PCF8574 I2C chip that converts I2C serial data to parallel data for the LCD display. There are many examples on the internet for using this board with Arduino. Do a search for “Arduino LCD PCF8574“. The I2C address is 0x3F

by default, but this can be changed via 3 solder jumpers provided on the board. This allows up to 3 LCD displays to be controlled via a single I2C bus (giving each one it's own address).

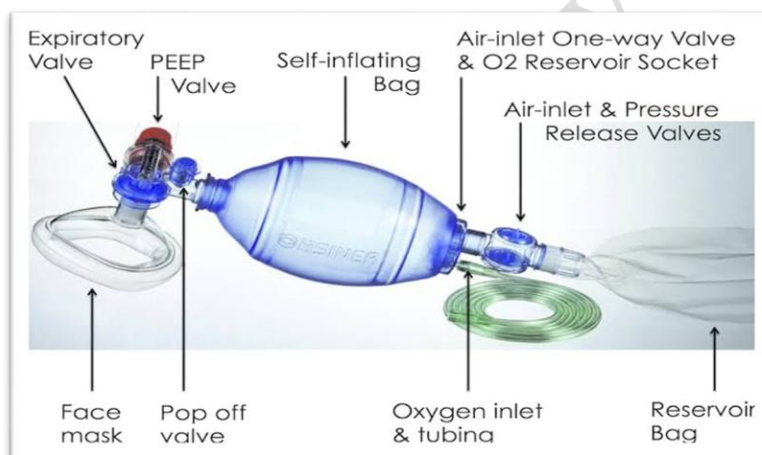
Stepper motor:



A 12-24V, 4.2kg-cm stepper motor is used in this ventilator for the proper working of the two arms. The stepper motor sets the starting position corresponding to the outmost clamp position as the reference. According to the initial set tidal volume and also the real-time deviation of

actual delivered tidal volume in subsequent cycles (determined from the feedback loop compensation) within a specified tolerance consists of a control circuit that provides feedback on the current position of the motor shaft, a feedback signal generated by comparing output signal and reference input signal. The programmable stepper motor drives lead screw-based Mechanism for smooth compression of the AMBU bag and, for delivery of the required amount Of volume. The picture below shows the portable ventilator and the graph between the strokeLength and the volume of air delivered by the ventilator. The device is portable with smooth Delivery of air/oxygen and can run on battery up to five hours.

Ambu bag :



A bag valve mask (BVM), sometimes referred to as an Ambu bag, is a handheld tool that is used to deliver positive pressure ventilation to any subject with insufficient or ineffective breaths. It consists of a self-inflating bag, one-way valve, mask, and an oxygen reservoir. The rescuer should be at the patient's head. A good seal must be achieved with the mask and the face. The pointed end of the mask must be over the nose, and the curved end just below the lower lip. Use of manual resuscitators to ventilate a patient is frequently called "bagging" the patient and is regularly necessary in medical emergencies when the patient's breathing is

insufficient (respiratory failure) or has ceased completely (respiratory arrest). Use of the manual resuscitator force-feeds air or oxygen into the lungs in order to inflate them under pressure, thus constituting a means to manually provide positive-pressure ventilation. It is used by professional rescuers in preference to mouth-to-mouth ventilation, either directly or through an adjunct such as a pocket mask.

16*2 LCD Display:



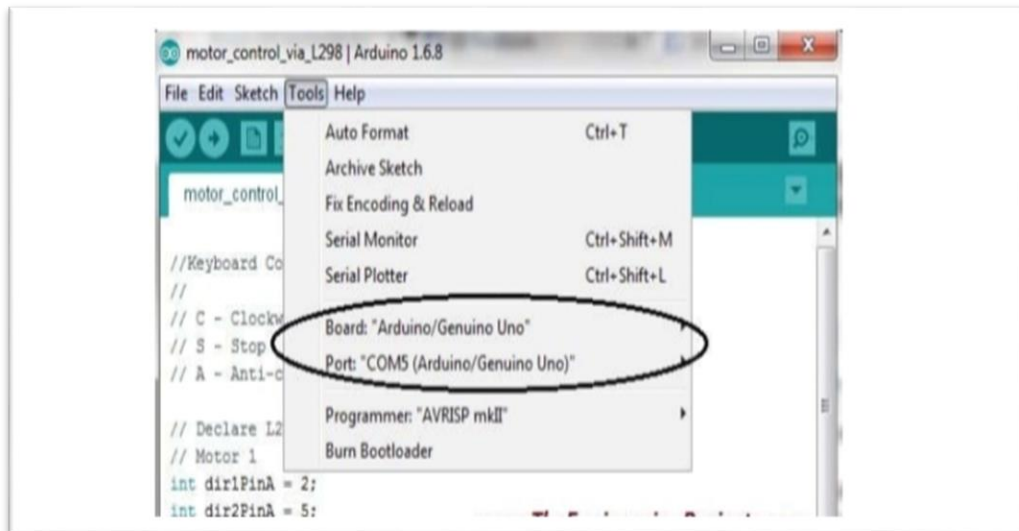
LCD Module LCD stands for liquid Display. The LCD screen is an alphanumeric display and it's various applications in several fields. This display might be a very essential module and is most normally employed in devices and circuits. A 16 x 2 LCD means it will be wont to display a maximum of 16 characters per line, and there are two such lines. Each character through this LCD is displayed in a very 5x7 pixel matrix format. The digital display is capable to display 224 various characters and symbols in two modes 4-bit and 8-bit. It consists of 16 pins. this may be operated

between 4.7 V to 5.3 V.

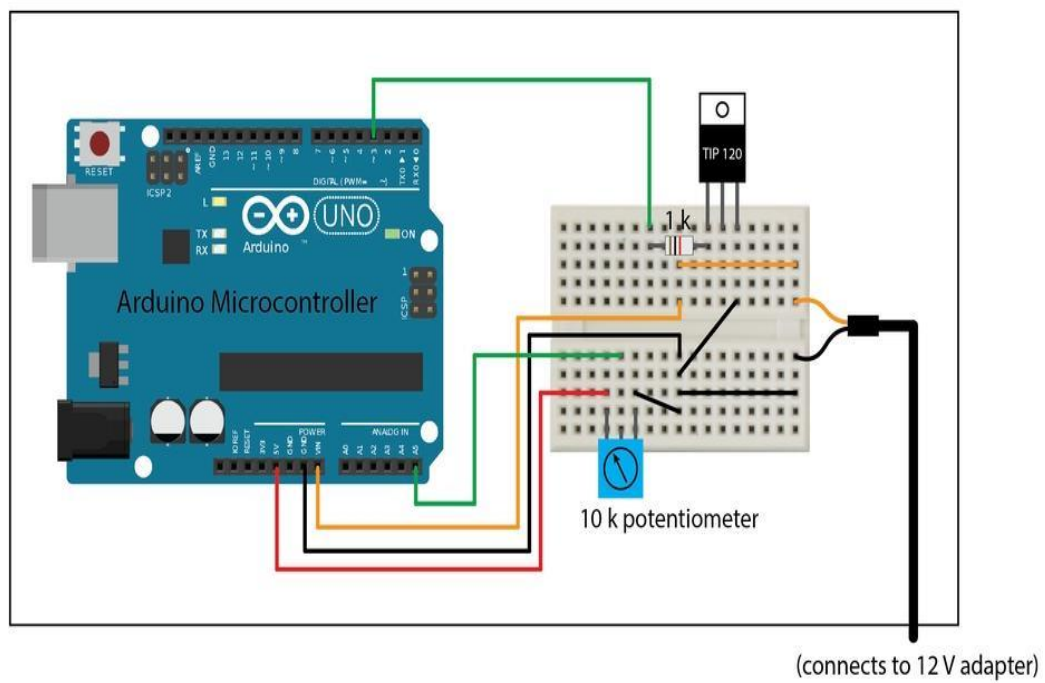
Software Implementation

Arduino Code Designing:

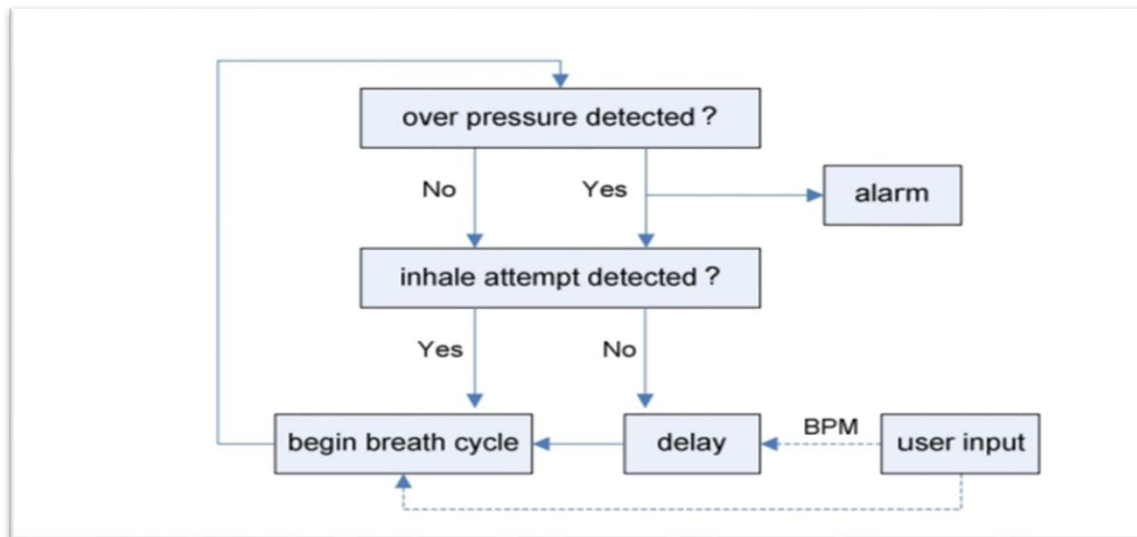
- ☐ Open the Arduino IDE software and go to Tools and select the Arduino board and the COM port properly.
- ☐ Now, upload the source code for 2 sec clockwise and 3 sec anti clockwise rotation to the Arduino UNO Board. So that, it can complete 12 RR/ min as per our calculation. Code is also available in Arduino IDE Website. Just past the code and edit the time value, connection pin number and motor speed set carefully.



3.3. Circuit Diagram



3.4 Flow chart



Arduino required voltage is 5V DC supply. A LCD show is utilized for show the message and it likewise required 5V DC Supply. Arduino are required three essential need pr supply, Reset circuit and oscillator unit. The ventilator we here plan And foster utilizing Arduino envelops of those prerequisites to Create solid yet reasonable DIY ventilator to help in the midst pandemic.. Our framework utilizes blood oxygen sensor Alongside delicate strain sensor to notice the predetermined Vitals of the patent and show on a small screen. Additionally a Crisis ringer alert is fitted inside the framework to sound a Ready when any abnormality is identified. The whole Framework is driven by Arduino regulator to appreciate Wanted results and to help patients in COVID pandemic and Other crisis circumstances.

Chapter 4

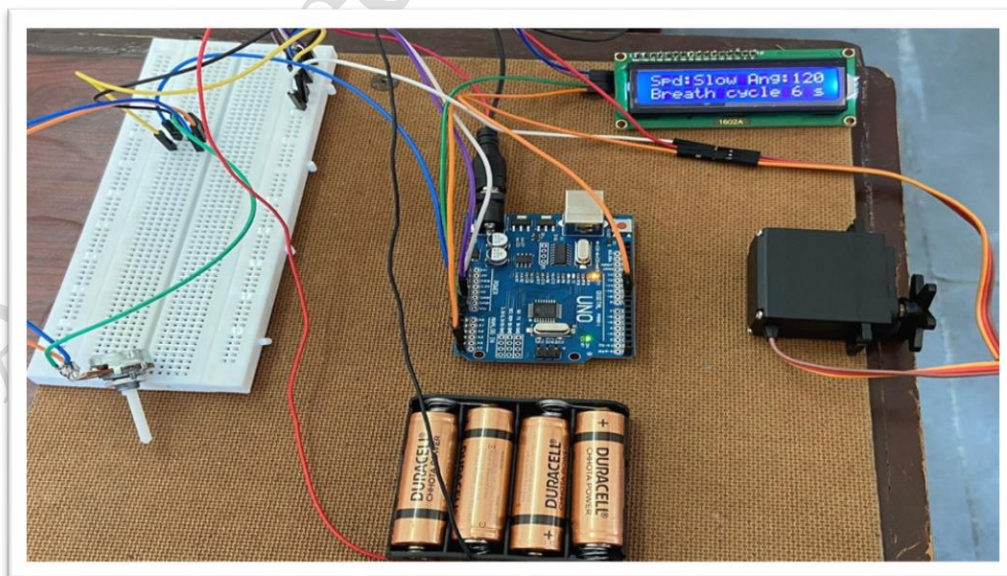
RESULTS

After implementing of all setup and successful run of Arduino code operated several time to obtain result.

□ According to our uploaded source code motor will rotate 2 sec clockwise and 3 sec anti clockwise continuously. So that, it can complete 12 Respiratory rate (RR)/min as per our calculation to provide a continuous plate pressure and air flow for the pneumonia cases of COVID-19 patient lungs.

Designed and developed a prototype of automatic ventilator to support pneumonia cases ,all sorts of respiratory patients.

- It features assist control and provide a constant air flow to the lungs.
- Power requirement is very low and running for 3.5 hours on one battery charge at its most demanding setting. □ It is low cost, portable, light weight and able to run 2 DC motors at the same time.
- Further development includes pressure sensor, over pressure alarm and other safety features to make it more user friendly



The above figure shows the LCD screen showing the breath cycle which means breathing rate by counting the number of breaths you take over the course of one minute while you're at rest.

Chapter 5

5.1. Conclusions

The lack of adequate ventilator support has already caused deaths in the first and second wave of COVID-19 pandemic and more can be expected unless ventilators can quickly be provided to areas overburdened with respiratory patients both now and in the inevitable future surges of infection. The ventilator using AMBU bag, developed in this project, can be life saving unit during medical emergencies. Emergency oxygen supply will be provided until patient receives any medical aid. The gear motor used is made to push the AMBU bag to create air pressure, simultaneously the heart pulse and oxygen saturation level of the patient is monitored. The developed portable ventilator can be operated by any less skilled person in case of medical emergency.

This work is a reasonable strategy potential for crisis and Covid pandemic. It is an open source ventilator configuration created utilizing disseminated fabricating. This paper is an itemized clarification of delivering minimal expense, open source mechanical ventilators for patients. Sure this work will acquire more noteworthy consideration. There is a great deal of future work to be moved up to make it clinical grade equipment. It is a major hotspot for both the ongoing pandemic circumstance and crisis purposes and in any event, for regular use in low asset settings.

5.2. Future scope

In this project, a prototype device to assist the patients who can partially breathe by their own is developed.

We can involve this venture in season of crisis as a first help device. For instance: If an individual gets a respiratory issue. He really wants to taken to clinic quickly while going in rescue vehicle or in the mishap area he wanted of ventilator to breath so around then our venture is little, helpful and to work which can save a daily existence. Since the cost of our project is reasonable it is simple purchase by a destitute group to rich individuals. We add BP sensor and pressure sensor for more exactness.

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- Design and Prototyping of a Low–cost portable Mechanical Ventilator”, coined by Abdul Mohsen Al Hussein, Heon Ju Lee, Justin Negrete, Stephen Powelson, Amelia Servi, Alexander Slocum, Jussi Saukkonen, IEEE dated in 2010, USA
- MIT Emergency ventilator design Toolbox

SAMPLE CODE

```
#include <Servo.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
// Set the LCD address to 0x27 for a 16 chars and 2 line display
LiquidCrystal_I2C lcd(0x27, 16, 2);

Servo myservo; // create servo object to control a servo

int potpin = 0; // analog pin used to connect the potentiometer
int val; // variable to read the value from the analog pin
float pos = 0;
void setup() {

  lcd.begin();
  // Turn on the backlight and print a message.
  lcd. Backlight();
  myservo.attach(9); // attaches the servo on pin 9 to the servo object
  Serial.begin(9600);
```

Approved by HOD-ECE

```

    lcd.setCursor(0,0);      //sets the cursor at row 0 column 0
    lcd.print("Emergency Vent"); // prints 16x2 LCD MODULE
    lcd.setCursor(2,1);      //sets the cursor at row 1 column 2
    lcd.print("B8 MINIPROJECT")
    delay(4000);
}

void loop() {
    val = analogRead(potpin);      // reads the value of the potentiometer (value between 0 and
    1023)
    val = map(val, 0, 1023, 0, 180); // scale it to use it with the servo (value between 0 and 180)
    myservo.write(val); // sets the servo position according to the scaled value
    Serial.println(val);
    delay(15); // waits for the servo to get there

    if (val <=30 ) {
        lcd.setCursor(0,0);      //sets the cursor at row 0 column 0
        lcd.print("Spd:Fast Ang:100 "); // prints 16x2 LCD MODULE
        lcd.setCursor(0,1);      //sets the cursor at row 1 column 2
        lcd.print("Breath cycle 4 sec ");
        for (pos = 0; pos <= 100; pos += 1) { // goes from 0 degrees to 180 degrees
            // in steps of 1 degree
            myservo.write(pos);      // tell servo to go to position in variable 'pos'
            delay(15);
        }
        for (pos = 100; pos >= 0; pos -= 1) { // goes from 180 degrees to 0 degrees
            myservo.write(pos);      // tell servo to go to position in variable 'pos'
            delay(15); // waits 15ms for the servo to reach the position
        }
    }

    else if (val >=31 && val<=60 ) {

```

```

    lcd.setCursor(0,0);      //sets the cursor at row 0 column 0
    lcd.print("Spd:Fast Ang:110 "); // prints 16x2 LCD MODULE
    lcd.setCursor(0,1);      //sets the cursor at row 1 column 2
    lcd.print("Breath cycle 4.43 sec ");
    for (pos = 0; pos <= 110; pos += 1) { // goes from 0 degrees to 180 degrees
        // in steps of 1 degree
        myservo.write(pos);      // tell servo to go to position in variable 'pos'
        delay(15);
    }
    for (pos = 110; pos >= 0; pos -= 1) { // goes from 180 degrees to 0 degrees
        myservo.write(pos);      // tell servo to go to position in variable 'pos'
        delay(15); // waits 15ms for the servo to reach the position
    }

    else if (val >=61 && val<=90 ) {
        lcd.setCursor(0,0);      //sets the cursor at row 0 column 0
        lcd.print("Spd:Fast Ang:120 "); // prints 16x2 LCD MODULE
        lcd.setCursor(0,1);      //sets the cursor at row 1 column 2
        lcd.print("Breath cycle 3.53 sec ");
        for (pos = 0; pos <= 120; pos += 1) { // goes from 0 degrees to 180 degrees
            // in steps of 1 degree
            myservo.write(pos);      // tell servo to go to position in variable 'pos'
            delay(15);
        }
        for (pos = 120; pos >= 0; pos -= 1) { // goes from 180 degrees to 0 degrees
            myservo.write(pos);      // tell servo to go to position in variable 'pos'
            delay(15);
        }

    }

    else if (val >=91 && val<=120 ) {
        lcd.setCursor(0,0);      //sets the cursor at row 0 column 0
        lcd.print("Spd:Slow Ang:100 "); // prints 16x2 LCD MODULE
        lcd.setCursor(0,1);      //sets the cursor at row 1 column 2

```

```

lcd.print("Breath cycle 5 sec ");
for (pos = 0; pos <= 100; pos += 0.6) { // goes from 0 degrees to 180 degrees
  // in steps of 1 degree
  myservo.write(pos);          // tell servo to go to position in variable 'pos'
  delay(15);
}
for (pos = 100; pos >= 0; pos -= 0.6) { // goes from 180 degrees to 0 degrees
  myservo.write(pos);          // tell servo to go to position in variable 'pos'
  delay(15);
  /// / waits 15ms for the servo to reach the position
}
}

```

```

else if (val >=121 && val<=150 ) {
  lcd.setCursor(0,0);          //sets the cursor at row 0 column 0
  lcd.print("Spd:Slow Ang:110 "); // prints 16x2 LCD MODULE
  lcd.setCursor(0,1);          //sets the cursor at row 1 column 2
  lcd.print("Breath cycle 5.5 sec ");
  for (pos = 0; pos <= 110; pos += 0.6) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    myservo.write(pos);          // tell servo to go to position in variable 'pos'
    delay(15);
  }
  for (pos = 110; pos >= 0; pos -= 0.6) { // goes from 180 degrees to 0 degrees
    myservo.write(pos);          // tell servo to go to position in variable 'pos'
    delay(15);
    // waits 15ms for the servo to reach the position
  }
}

```

```

else if (val >=151 && val<=180 ) {

  lcd.setCursor(0,0);          //sets the cursor at row 0 column 0
  lcd.print("Spd:Slow Ang:120 "); // prints 16x2 LCD MODULE
  lcd.setCursor(0,1);          //sets the cursor at row 1 column 2
  lcd.print("Breath cycle 6 sec ");

```

```
for (pos = 0; pos <= 120; pos += 0.6) { // goes from 0 degrees to 180 degrees
  // in steps of 1 degree
  myservo.write(pos);          // tell servo to go to position in variable 'pos'
  delay(15);
}
for (pos = 120; pos >= 0; pos -= 0.6) { // goes from 180 degrees to 0 degrees
  myservo.write(pos);          // tell servo to go to position in variable 'pos'
  delay(15);
  // waits 15ms for the servo to reach the position
}
}
```

Approved by HOD-ECE